DESIGN SYSTEMS

TECHNICAL DATA SHEET MICRO-TOPP
M / BS / FINE – SULPHATE-BASED MINERAL COATING

TECHNICAL CHARACTERISTICS

MICRO-TOPP is the only coating on the market based on a sulphate-based mineral binder with the following performance: mechanical performance superior to any Portland cement; zero efflorescence under all conditions since, unlike all other mineral binders, it contains no free lime; rapid setting and high early strengths, with one tenth of the shrinkage of Portland cement and one fifth of the porosity; greater resistance to sulphates, acids and all aggressive environments; a low pH value, which makes it more compatible with all the polyurethane protective coatings on the market.

Thanks to its special formulation, MICRO-TOPP makes it possible to obtain different types of finish, all with pleasant, modern aesthetic characteristics. Its easy application and reliable preparation method allow the installer to guarantee consistent performance and aesthetic effects, regardless of the application technique used and the effect desired: monolithic, clouded, scratched, trowelled, glossy surfaces, etc., on substrates of various kinds. Mixed with water, it has a light, creamy workability and is easily trowelled in all its grain sizes, on a wide variety of surfaces, including over glass-fibre mesh. Its very fast drying makes it sandable, where required, after about 4 hours.

- For exterior and interior use, on walls and floors.
- Can be applied at temperatures from 5°C to 35°C.
- Available in 25 kg and 14 kg packs.
- Available in various grain sizes.
- Coloured.
- Mechanical mixing.
- Application by trowel or stainless steel/plastic float.
- Very high mechanical performance.
- No efflorescence.
- Rapid setting.
- Resistance to sulphates and acids.
- Very low CO2 emissions.
- Easy to apply, cement-free.
- Use personal protective equipment.

USES

Designed mainly for decorating floors, walls and furnishings, it is excellent for creating surfaces with a textured look in interior and exterior environments. Ideal for commercial and residential use, it guarantees excellent aesthetic quality; it can also be used as a finish for swimming pools, after waterproofing with suitable products.

SUBSTRATES

It allows the most varied surfaces to be finished: wood panels, MDF, ceramic, plasterboard, fibre cement, plywood.

DO NOT USE

On non-waterproofed swimming pools, on rubber, linoleum, polyurethane foam, unprimed plasterboard, non-cementitious waterproofing membranes and in the presence of rising damp.

APPLICATION

Before applying the product, check that the substrate is compatible, sound, cured and free from dust, oil, grease, etc. If necessary, add the colourant to obtain the desired shade.

Mix the product with clean water, pouring it into the powder in the prescribed quantity without exceeding the indicated percentage, and mix with a low-speed helical mixer until a homogeneous, lump-free paste is obtained. Let the product rest for 2-3 minutes before application, to ensure optimal hydration of all the components. Application must be carried out with a stainless steel or plastic trowel or float.

TYPICAL PRODUCT DATA

DESCRIPTION	RESULT
Colour / consistency	Coloured / powder
Solids content	100%
Bulk density	Approx. 1,55 kg/l
Hazard classification (Directive 1999/45/EC)	Irritant: read carefully the instructions given in the safety data sheet
Storage	12 months in a dry, ventilated place
Packaging	14 kg
Customs tariff code	3824 50 90

VALUES AND APPLICATION INFORMATION (AT 23°C AND 50% R.H.)

DESCRIPTION	RESULT
Mixing water	27% to 37% water per one part of MICRO-TOPP M (depending on the colour and effect required)
Working time of the mix	Approx. 50 minutes
Consumption	Approx. 1,5 kg/m^2 per mm of thickness
Mineral content	> 35%
Free calcium hydroxide	None

Application temperature	From +20°C to +40°C (not in full sun)

DESCRIPTION	RESULT
Time between two coats	Also wet-on-wet (recommended to maximise the textured effect)
Time before finishing	6 hours
Ready for use	Depends on the finish
pH of the mix	Approx. 11

FINAL PERFORMANCE VALUES

DESCRIPTION	RESULT
Compressive strength (grain size 0-200 µm, 28 days)	> 35 MPa
Flexural strength (grain size 0-200 µm, 28 days)	> 7 MPa
Abrasion resistance (grain size 0-200 µm, 28 days, 3.000 revolutions – UNI EN ISO 5470-1)	< 1 g
Capillary water absorption coefficient (EN 1015-18)	c < 20 kg/(m² · min^0,5)
Water vapour permeability (EN 1745)	µ < 25
Reaction to fire	Class F
Taber abrasion (S42 strips, 1.000 g, 100 revolutions)	0,1 g
Taber abrasion (S42 strips, 1.000 g, 300 revolutions)	0,2 g
Taber abrasion (S42 strips, 1.000 g, 500 revolutions)	0,3 g
Taber abrasion (S42 strips, 1.000 g, 1.000 revolutions)	0,3 g

The technical data given in this bulletin were obtained using standardised tests and standards, or modifications thereof, according to Design Systems systems. The methods applied may be requested from our technical service. The contents of this technical bulletin represent the best of our knowledge and experience. However, for uses other than those indicated, preliminary tests are recommended. In any case, Design Systems Srl accepts no responsibility for damage or defects caused by the use of its products, since the conditions of use are not under its control. It also advises that its technical service is available to customers for any information concerning the correct use of its products.